NEUROMORPHIC COMPUTING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Neuromorphic Computing

Question 1

10.0 points

Examine the diagram below of a leaky integrate-and-fire (LIF) neuron model, which shows the membrane potential trace over time. Analyze how the membrane potential changes in response to incoming spikes and explain the conditions under which the neuron fires. Specify any key parameters illustrated in the diagram, such as threshold potential and leakage rate.

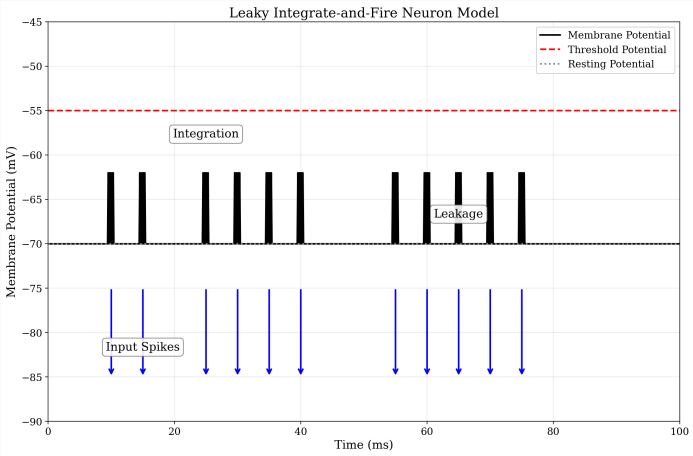

Fig. 1. CUMULATIVE LEAK CORRECTIONS (all must be fixed): The diagram contains labels "Integration" and "Leakage; The diagram contains the labels "Integration" and. A membrane potential trace for a leaky integrate-and-fire (LIF) neuron model, showing how the membrane potential changes over time in response to incoming spikes. The diagram includes key parameters such as the threshold potential and leakage rate, with labeled axes for time and membrane potential. The trace should depict the potential rising with incoming spikes, leaking over time, and resetting after reaching the threshold. IMPORTANT correction needed: The diagram contains labels "Integration" and "Leakage. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT correction needed: The diagram contains the labels "Integration" and. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 2 10.0 points

Examine the diagram below illustrating a spike-timing dependent plasticity (STDP) learning rule. Describe how the timing of pre- and postsynaptic spikes affects the synaptic weight change. Discuss the implications of synaptic modifications based on different time intervals between spikes.

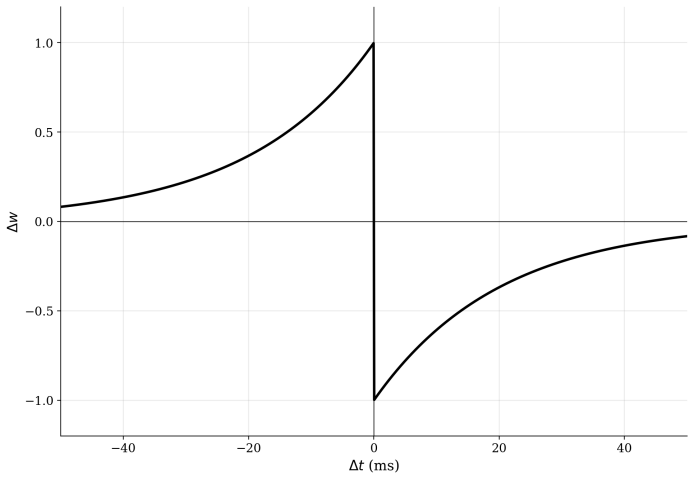

Fig. 2. A diagram illustrating the spike-timing dependent plasticity (STDP) curve, showing synaptic weight change as a function of the time difference between pre- and postsynaptic spikes. The x-axis represents the time difference (Δt) between spikes, and the y-axis represents the change in synaptic weight. The curve typically shows potentiation (positive weight change) for positive Δt and depression (negative weight change) for negative Δt, highlighting the effect of spike timing on synaptic modifications. IMPORTANT: The diagram is missing required labels. The diagram is missing labels for the potentiation and. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows an anti-Hebbian ST. Regenerate using EXACTLY the values and sequences specified.

Question 3

10.0 points

Examine the diagram below of a crossbar array architecture used for storing synaptic weights in neuromorphic computing. Explain how this architecture enables efficient weight storage and retrieval. Highlight any challenges related to scaling and variability illustrated in the diagram.

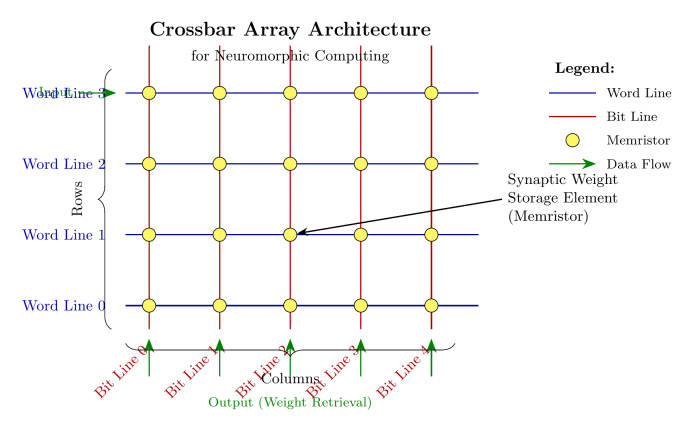

Fig. 3. A crossbar array architecture used in neuromorphic computing, showing a grid of horizontal and vertical lines representing rows and columns. Each intersection has a synaptic weight storage element, such as a memristor. Highlight the challenges related to scaling, such as increased line resistance and variability in device characteristics. Include labels for rows, columns, and synaptic weights, and indicate potential issues with scaling and variability. IMPORTANT: The diagram is missing required labels. The diagram description requires indicating potential issues with scaling. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description requires a label for "synaptic weights. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 4

10.0 points

Evaluate the temporal coding scheme in the spiking neural network diagram below. Describe how information is encoded temporally and its impact on network processing capabilities. What are the advantages of temporal coding compared to rate coding based on the diagram?

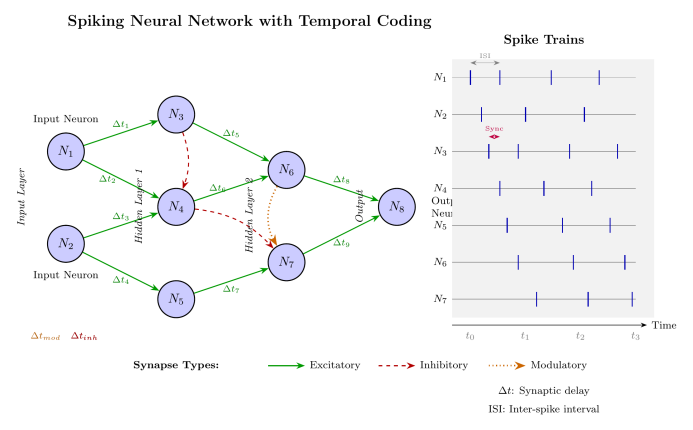

Fig. 4. A spiking neural network diagram showing a set of neurons with labeled spike timings. Each neuron is represented as a node, and spikes are shown as vertical lines on a timeline for each neuron. The diagram should illustrate how temporal coding is used to encode information, with arrows indicating the flow of spikes between neurons. Include labels for neurons and time intervals to highlight the temporal aspect of the coding scheme. IMPORTANT: The diagram is missing required labels. The diagram is missing several required labels for the. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description explicitly. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 5 10.0 points

Examine the comparative diagram of energy consumption between von Neumann and neuromorphic architectures shown in the diagram below. Discuss the reasons for differences in energy efficiency. What implications does this have for large-scale neural applications?

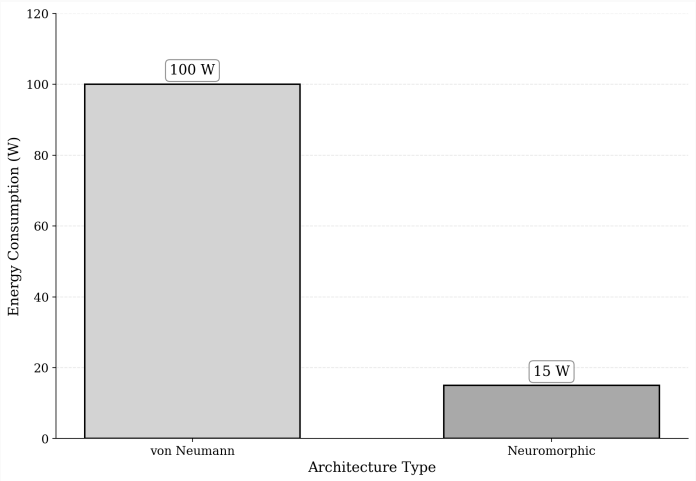

Fig. 5. A comparative energy consumption chart showing energy usage of von Neumann vs. neuromorphic architectures. The chart should have two bars or lines representing each architecture, with energy consumption on the y-axis and architecture type on the x-axis. Include labels for each architecture and a title indicating the comparison of energy consumption.